A legume-based hypocaloric diet reduces proinflammatory status and improves metabolic features in overweight/obese subjects.

BACKGROUND: The nutritional composition of the dietary intake could produce specific effects on metabolic variables and inflammatory marker concentrations. This study assessed the effects of two hypocaloric diets (legume-restricted- vs. legume-based diet) on metabolic and inflammatory changes, accompanying weight loss. METHODS: Thirty obese subjects (17 M/13F; BMI: 32.5 ± 4.5 kg/m(2); 36 ± 8 years) were randomly assigned to one of the following hypocaloric treatments (8 weeks): Calorie-restricted legume-free diet (Control: C-diet) or calorie-restricted legume-based diet (L-diet), prescribing 4 weekly different cooked-servings (160-235 g) of lentils, chickpeas, peas or beans. Body composition, blood pressure (BP), blood biochemical and inflammatory marker concentrations as well as dietary intake were measured at baseline and after the nutritional intervention. RESULTS: The L-diet achieved a greater body weight loss, when compared to the C-diet (-7.8 ± 2.9% vs. -5.3 ± 2.7%; $p = 0.024$). Total and LDL cholesterol levels and systolic BP were improved only when consuming the L-diet ($p < 0.05$). L-diet also resulted in a significant higher reduction in C-reactive protein (CRP) and complement C3 (C3) concentrations ($p < 0.05$), compared to baseline and C-diet values. Interestingly, the reduction in the concentrations of CRP and C3 remained significantly higher to L-diet group, after adjusting by weight loss ($p < 0.05$). In addition, the reduction (%) in CRP concentrations was positively associated with decreases (%) in systolic BP and total cholesterol concentration specifically in the L-diet group, independent from weight loss ($p < 0.05$). CONCLUSION: The consumption of legumes (4 servings/week) within a hypocaloric diet resulted in a specific reduction in proinflammatory markers, such as CRP and C3 and a clinically significant improvement of some metabolic features (lipid profile and BP) in overweight/ obese subjects, which were in some cases independent from weight loss.